Low-Cost-UMI-Report

A Wireless Bimanual Data Collection System for Imitation Learning

1. Introduction

This report presents a **low-cost wireless bimanual data collection system** designed for imitation learning on the SO-101 robot arm. Each gripper unit captures 6DoF pose via T265 VIO, fisheye observation images, and gripper aperture from the STS3215 encoder — all logged onboard a Raspberry Pi 4B. Two grippers are synchronized via an **alternating peer-to-peer UDP protocol** over WiFi Direct, achieving zero cumulative timing error through symmetric time synchronization.

2. System Architecture

2.1 Hardware Architecture

2.2 Bill of Materials

2.3 Per-Gripper Software Architecture

Each RPi runs a single Python process with three threads feeding a synchronized buffer at 30 Hz. The architecture ensures that pose, image, and gripper data are temporally aligned before storage.

Feature	UMI	FastUMI	Ours
Observation	GoPro RGB	GoPro RGB	T265 fisheye
Pose tracking	ORB-SLAM3	T265 VIO	T265 VIO
Gripper width	Vision	ArUco	STS3215 encoder
Cameras / gripper	1	2	**1**
Compute	PC offline	Laptop + ROS	**RPi onboard**
Tethered	No	USB cable	**None**

Table 1: System comparison. Ours uses the fewest components while remaining fully wireless.

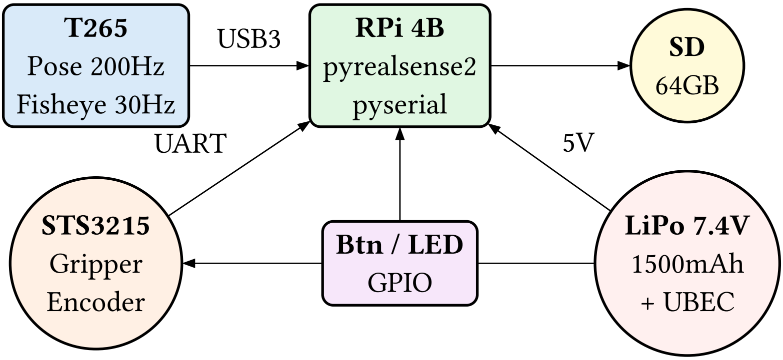

Figure 1: Single gripper unit hardware architecture. Total mass 340g, battery life 45 min.

Connection	From	To	Notes
USB 3.0	T265 Camera	RPi USB3 (blue)	Cable < 30cm
UART TX	RPi GPIO14	STS3215 RX	3.3V TTL
UART RX	RPi GPIO15	STS3215 TX	3.3V TTL
Power 7.4V	LiPo	STS3215 VCC	Direct connection
Power 5V	UBEC out	RPi USB-C	5V 3A rated
Button	Switch	GPIO17	Pull-up, active low
LED	GPIO27	LED + resistor	330Ω series

Table 2: Connection details for single gripper unit.

Component	Qty	Cost	Note
Intel RealSense T265	× 2	~$200 ea	eBay / AliExpress
Raspberry Pi 4B 4GB	× 2	~$55 ea	
MicroSD 64GB A2	× 2	~$12 ea	High write speed
Feetech STS3215	× 2	~$15 ea	12-bit encoder
7.4V 2S LiPo 1500mAh	× 2	~$15 ea	XT30 + BMS
5V 3A UBEC	× 2	~$5 ea	
Misc + 3D housing	× 2	~$20 ea	PLA + TPU
Bimanual total		**~$640**	

Table 3: BOM for two grippers. T265 price varies $150–250 on secondary market.

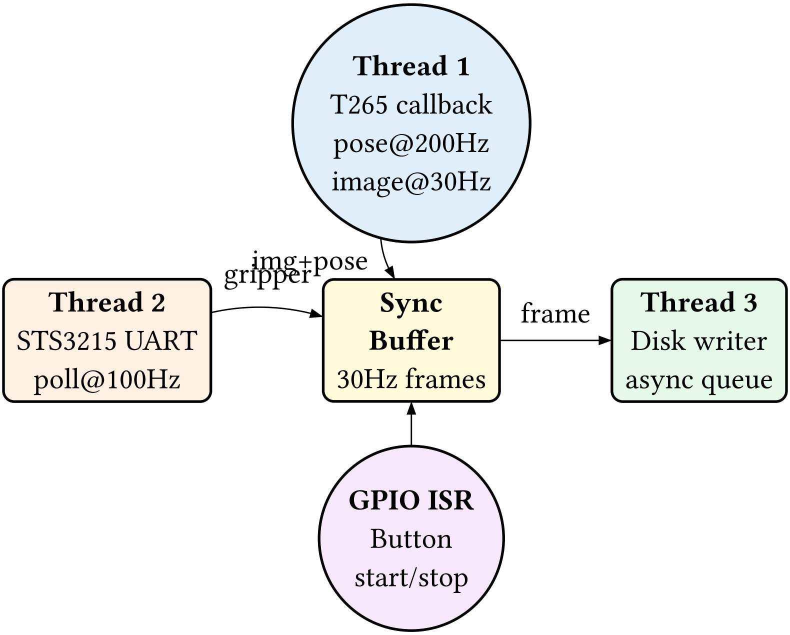

Figure 2: Per-RPi software threads. The T265 callback buffers poses at 200Hz and matches them to 30Hz fisheye frames (max error ± 2.5ms).

VIO drift is acceptable for our use case: at 0.3 m/s hand speed and 30 s episodes, drift stays below 5 mm. Imitation learning policies (Diffusion Policy, ACT) learn **relative** deltas — per-step drift (5µm) is negligible.

3. Bimanual Synchronization

3.1 Problem: Independent Loops Accumulate Error

If each RPi runs

```
sleep(1/30)
```

independently, Linux scheduler jitter (4µs/step) accumulates:

$$\Delta t(n) = \sum_{i=1}^{n} \left(T_A^i - T_B^i\right) \approx n \cdot 4\mu s$$

After 5 minutes (9000 steps): **36 ms drift** — more than one full 33 ms frame.

3.2 Solution: Alternating Peer-to-Peer Protocol

Duration	Steps	Indep.	UDP
3 s	100	0.4 ms	2 ms
30 s	1k	4 ms	2 ms
100 s	3k	12 ms	2 ms
5 min	9k	36 ms	2 ms

Table 4: Error comparison. Independent loop grows linearly; UDP trigger stays constant.

Unlike traditional master-slave architectures, our system uses symmetric peers that alternate trigger responsibilities. This provides balanced latency and eliminates single points of failure.

$$\Delta t(n) = \tau_{\mathrm{WiFi}} \approx 2 \text{ ms} \quad \forall n$$

3.3 Bimanual System Topology

3.4 Time Synchronization Phase

Before recording begins, both peers exchange timing information to establish clock offset estimates:

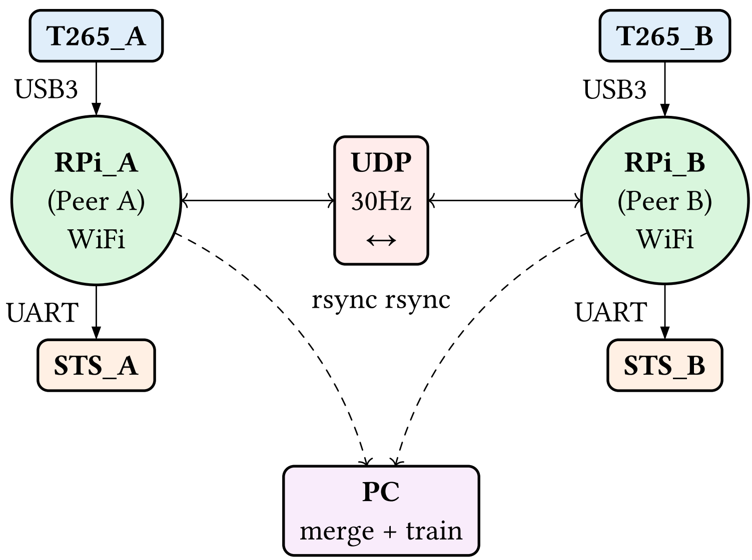

Figure 3: Bimanual topology with symmetric peers. Both grippers communicate bidirectionally via UDP.

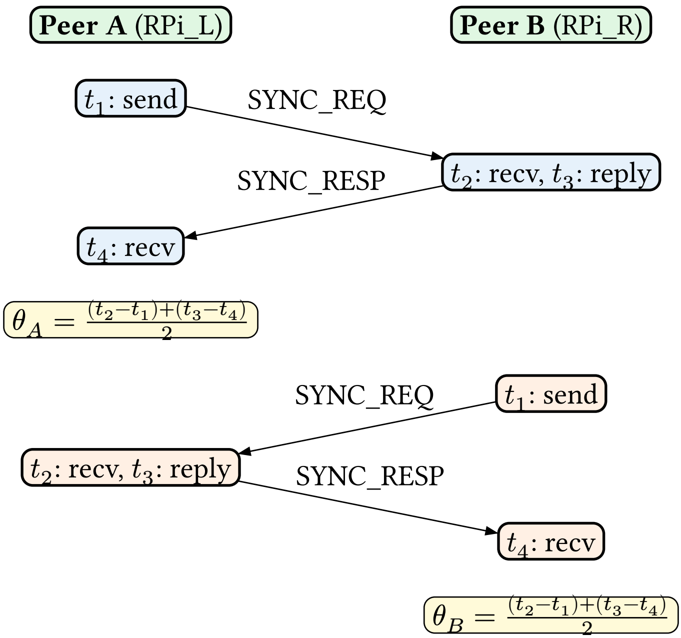

Figure 4: Bidirectional time synchronization. Each peer computes its clock offset using NTP-style round-trip measurement ($t_1 \ldots t_4$). Round 1 (blue) computes θ_A, Round 2 (orange) computes θ_B.

3.5 Recording Phase: Alternating Triggers

During recording, peers alternate sending step triggers, ensuring symmetric latency distribution:

3.6 Protocol Advantages

The alternating peer-to-peer design offers several key benefits:

- **Symmetric latency:** Both peers experience identical average network delay
- **No single point of failure:** Either peer can detect and recover from the other's failure
- **Fault tolerance:** If one peer misses a trigger, the other can take over
- **Balanced load:** Network and compute overhead is evenly distributed

3.7 Packet Format

Each trigger is 28 bytes, broadcast over UDP port 7265:

3.8 Timing Waveform

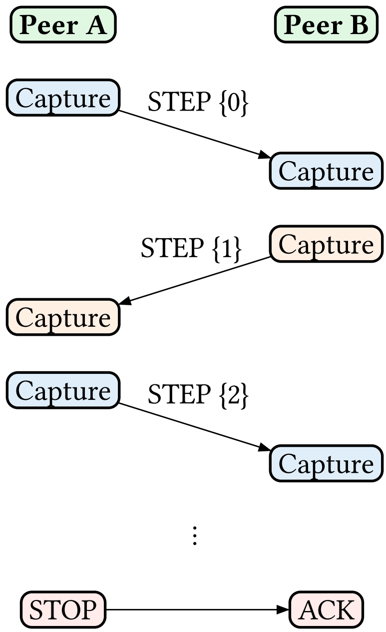

Figure 5: Alternating trigger protocol. Even steps (blue): Peer A captures first and sends trigger. Odd steps (orange): Peer B captures first and sends trigger. This ensures symmetric average latency.

Field	Type	Size	Role
magic	char[2]	2B	"GS" identifier
msg_type	uint8	1B	SYNC_REQ/SYNC_RESP/STEP/STOP
step	uint32	4B	Absolute step #
episode	uint16	2B	Episode index
flags	uint8	1B	REC / PAUSE
sender_id	uint8	1B	Peer A=0, B=1
timestamp	float64	8B	Sender clock (s)
sync_t2	float64	8B	For SYNC_RESP only
checksum	uint8	1B	XOR payload
		28B	

Table 5: UDP packet format. The absolute step number and sender_id prevent misalignment after packet loss.

	0	1	2	3	4	5	6	7
A capture	✓	✓			✓	✓		
A→ B		→	→			→	→	
B capture			✓	✓		✓	✓	
B→ A				←	←			←
Step		0		1		2		

Table 6: Timing diagram showing alternating triggers. Peer A captures and sends even steps (blue), Peer B captures and sends odd steps (orange). WiFi latency 1–3ms.

4. Data Pipeline

4.1 Dataset Storage (LeRobot v3.0 format)

Each gripper produces episodes stored in LeRobot v3.0 dataset format. Multiple episodes are packed into consolidated files, reducing filesystem overhead and enabling efficient streaming. After rsync to PC, episodes from both peers are merged into a single dataset.

At 30Hz with MP4-compressed fisheye video, sustained write is well within SD A2 spec.

```
dataset_name/
├── meta/
│   ├── info.json          # fps, robot_type, features, path templates
│   ├── stats.json         # global feature statistics (mean/std/min/max)
│   ├── tasks.jsonl         # task descriptions
│   └── episodes/
│       └── chunk-000/
│           └── file-000.parquet   # episode metadata (lengths, offsets)
├── data/
│   └── chunk-000/
│       └── file-000.parquet       # multi-episode frames
└── videos/
    └── observation.images.fisheye/
        └── chunk-000/
            └── file-000.mp4       # multi-episode video
```

Listing 1: LeRobot v3.0 dataset directory structure. Multiple episodes are packed into single data and video files.

Column	Type	Shape	Description
observation.state	float32	(10,)	Position (3) + quaternion (4) + gripper (1) + confidence (1) + side (1)
observation.images.fisheye	video	(800, 848, 1)	T265 fisheye frame (MP4)
action	float32	(10,)	Target state for next step
timestamp	float64		Seconds from episode start
episode_index	int64		Episode identifier
frame_index	int64		Frame position within episode
index	int64		Global frame index across dataset
task_index	int64		Links to tasks.jsonl

Table 7: Parquet columns per frame. Episode boundaries are resolved via meta/episodes/ metadata, not filenames.

4.2 Post-Collection: Merge and Train

Since both RPis share identical absolute step numbers, merging is a trivial zip-by-step operation:

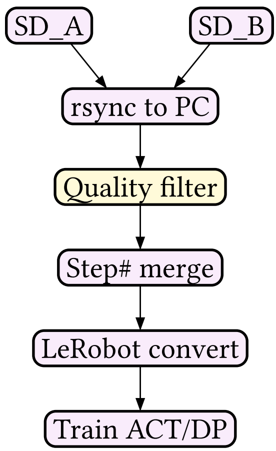

Figure 6: Post-collection pipeline with quality filtering. Episodes with confidence < 2 are automatically discarded.

5. TODO

This section lists the tasks required to build and validate the complete bimanual data collection system.

5.1 Procurement and Unit Tests

Goal: Confirm every component works individually.
□ Acquire 2 T265 cameras (eBay / AliExpress). Verify firmware version 0.2.0.951.
□ Flash Ubuntu 22.04 Server 64-bit on both RPi 4Bs.
□ Build librealsense v2.53.1 from source with FORCE_RSUSB_BACKEND and Python bindings enabled.
□ Add T265 USB autosuspend udev rule (disable autosuspend for vendor 8087, product 0b37).
□ Enable GPIO UART on /dev/ttyAMA0. Verify STS3215 position read at 100Hz.
□ Verify pyrealsense2 imports successfully and rs-pose outputs 6DoF. Verify STS3215 returns encoder values.

5.2 Single Gripper Recorder

Goal: Record one gripper's data (pose + fisheye + gripper width) to SD at 30Hz.
□ Implement T265 callback: buffer pose at 200Hz, capture fisheye at 30Hz.
□ Implement STS3215 UART polling thread at 100Hz.
□ Build 3-way synchronizer: on each fisheye frame, match nearest pose (± 2.5ms) and nearest gripper reading (± 5ms).
□ Implement episode recorder with GPIO button (start/stop) and LED indicator.
□ Save episodes to SD card. Verify sustained 30Hz write with no dropped frames.
□ Build playback visualizer: fisheye + pose trajectory + gripper width overlay.
□ Validate: 30-second episode records cleanly with confidence ≥ 2 throughout.

5.3 UDP Bimanual Synchronization

Goal: Two grippers capture in lockstep via alternating UDP trigger.
□ Configure WiFi Direct connection between both RPis.
□ Implement 28-byte UDP packet format (msg_type, step, episode, flags, sender_id, timestamp, sync_t2, checksum).
□ Implement time synchronization phase with bidirectional SYNC_REQ / SYNC_RESP exchange.
□ Implement alternating trigger loop: even steps triggered by Peer A, odd steps by Peer B.
□ Implement packet-loss detection (step gap) with empty frame insertion.
□ Add periodic RTT measurement (every 300 steps) and latency logging.
□ Stress test: 10-minute continuous run. Verify A/B step counts match exactly.

□ Validate: average RTT < 5ms; zero step misalignment after 10 minutes.

5.4 Hardware Integration

Goal: Two complete wireless gripper units.
□ Design 3D housing in Fusion 360: T265 mount (forward-facing, TPU vibration pads), RPi slot (SD card accessible, heatsink clearance), LiPo bay (hot-swappable, XT30 connector), STS3215 parallel-jaw mechanism, ergonomic handle with trigger button.
□ Print (PLA frame + TPU fingertips), assemble, and wire both units.
□ Gripper calibration: STS3215 encoder count ⟶ mm aperture mapping.
□ Battery endurance test: confirm 45+ min per charge.

5.5 Pilot Data Collection

Goal: 50+ bimanual demonstration episodes.
□ Collect tabletop pick-and-place tasks (cups, blocks, tools).
□ Auto-quality filter: discard any episode where confidence drops below 2.
□ Transfer data to PC via rsync. Run merge script matching step numbers.
□ Verify merged dataset integrity: A/B frame counts match, no gaps.

5.6 Policy Training and Deployment

Goal: Close the loop from data to autonomous execution.
□ Convert merged episodes to LeRobot dataset format.
□ Train Diffusion Policy or ACT on fisheye observation + pose state.
□ Mount T265 + gripper on SO-101 follower arm wrist.
□ Evaluate autonomous bimanual task execution.

6. Risk Assessment

7. Conclusion

This report presented a low-cost wireless bimanual data collection system designed for imitation learning. Key contributions include:

1. **Fully wireless architecture:** Each gripper unit operates independently with onboard compute and battery, eliminating tethering constraints during data collection.

2. **Alternating peer-to-peer synchronization:** The UDP protocol ensures symmetric latency distribution and eliminates single points of failure, achieving bounded timing error regardless of recording duration.

3. **Step-based alignment:** Absolute step numbers enable trivial post-collection merging without complex timestamp alignment algorithms.

Risk	Level	Mitigation
T265 unavailable (discontinued)	Med	Buy 2+ units now; fallback: OAK-D Lite + DepthAI VIO
librealsense ARM64 build failure	Med	v2.53.1 + FORCE_RSUSB_BACKEND is verified on RPi4
Mono fisheye hurts policy	Med	Add small USB RGB cam to RPi USB2 port if needed
WiFi packet loss > 1%	Low	WiFi Direct (no router); auto-interpolation of gaps
RPi thermal throttle	Low	Heatsink + fan; 15-min sessions with battery swap
STS3215 UART unstable	Low	74HC126 buffer IC or USB-serial adapter fallback

Table 8: Key risks and mitigation strategies.

4. **Cost efficiency:** The complete bimanual system costs approximately $640, significantly less than alternatives requiring external tracking infrastructure.

Future work includes integrating the system with the SO-101 robot arm for closed-loop policy evaluation and exploring multi-gripper configurations beyond bimanual setups.